

Bergman beta heuristic

The goal of this short note is to provide a heuristic on why the limit(s) of finite-number-of-particles log gases in the disk are empty for $\beta = 4$.

Recall that we consider the Gibbs point process

$$\mathbb{P}_n(d\gamma) = \frac{1}{Z_{N,\beta}} e^{-\beta H(\gamma)} \text{Bin}_{\mathbb{D},n}(d\gamma),$$

where $\text{Bin}_{\mathbb{D},n}$ is the binomial point process in the unit disk $\mathbb{D}$ and

$$H(\gamma) = \sum_{\{x,y\} \subset \gamma} \log |x - y|.$$

For a class $\mathcal{C}$ of configurations such that, for all $\gamma \in \mathcal{C}$,

$$H(\gamma) \simeq \text{Cst},$$

the restricted free energy is given by

$$F_{n,\beta}(\mathcal{C}) := -\frac{1}{\beta} \log \int_{\mathcal{C}} e^{-\beta H(\gamma)} \text{Bin}_{\mathbb{D},n}(d\gamma) \simeq \text{Cst} - \frac{1}{\beta} \log \text{Bin}_{\mathbb{D},n}(\mathcal{C}).$$

This quantity allows us to compare how probable different classes of configurations are. We know (at least for $\beta = 2$), that almost all particles concentrate on the boundary of the disk. To see where the $\beta = 4$ comes from we will compare the asymptotics of candidates to be typical configuration:

1. The first candidate will be configurations where all the particles goes to the boundary leaving a limiting empty interior
2. The second candidate will be configurations with essentially all points on the boundary but one remains near 0

We will see that for $\beta > 4$ the free energy cost of candidates 1 will be favorable. This suggest empty point processes at the limit.

Let us define our two types of configurations: $\mathcal{C}_1$ and $\mathcal{C}_2$.

Let $n \geq 2$ and denote by

$$\Gamma_n := \{\gamma \subset \mathbb{D} : \#\gamma = n\}$$

the space of configurations with n points in the unit disk.

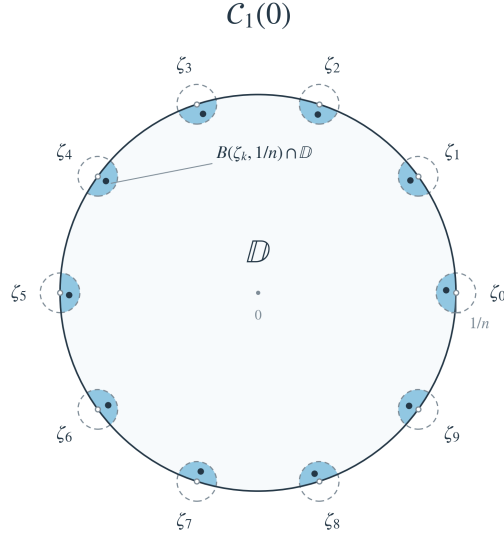
For $m \geq 1$, write

$$\zeta_k^{(m)} := e^{2\pi i k/m}, k = 0, \dots, m-1.$$

We define the reference class $\mathcal{C}_1$ by

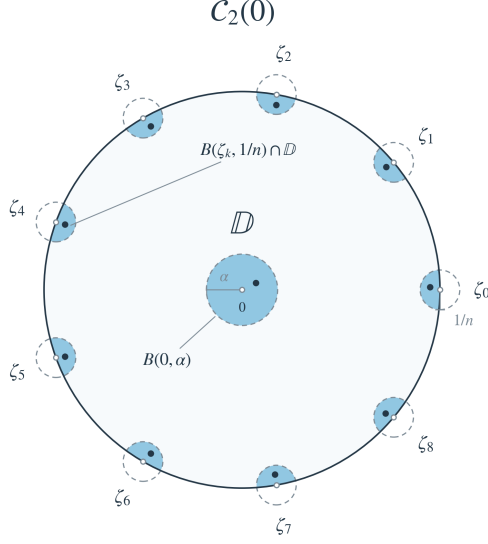
$$\mathcal{C}_1 := \left\{ \gamma \in \Gamma_n : \#(\gamma \cap B(\zeta_k^{(m)}, 1/n) \cap \mathbb{D}) = 1 \text{ for } k = 0, \dots, n-1 \right\}.$$

The figures illustrate $n = 10$. Black dots represent particles; hollow circles mark the centers of the balls.



Thus there is exactly one point in each blue boundary region, and no points elsewhere. We then consider

The boundary pattern may have any common orientation, while particles may vary within their respective regions.



Let $\alpha > 0$ be small. Similarly, we define the reference class $\mathcal{C}_2$ by

$$\mathcal{C}_2 := \left\{ \gamma \in \Gamma_n : \begin{array}{l} \#(\gamma \cap B(0, \sqrt{\alpha/\pi})) = 1, \\ \#(\gamma \cap B(\zeta_k^{(m)}, 1/n)) = 1 \text{ for } k = 0, \dots, n-2 \end{array} \right\}.$$

Thus there is exactly one point in the central ball and one point in each of the $n-1$ blue boundary regions, with no points elsewhere.

We now compute the binomial probabilities of these two classes. Write $|E|$ for planar area and set

$$q_n := \frac{|B(1, 1/n) \cap \mathbb{D}|}{\pi}.$$

Thus q_n is the contribution of one boundary cap. The central ball contributes α .

We then have

$$\text{Bin}_{\mathbb{D},n}(\mathcal{C}_1) = n! q_n^n, \quad \text{Bin}_{\mathbb{D},n}(\mathcal{C}_2) = n! \alpha^2 q_n^{n-1}.$$

We have

$$q_n \simeq \frac{1}{2n^2}.$$

Hence

$$\text{Bin}_{\mathbb{D},n}(\mathcal{C}_1) \simeq n! 2^{-n} n^{2n}, \quad \text{Bin}_{\mathbb{D},n}(\mathcal{C}_2) \simeq n! \alpha^2 2^{1-n} n^{2-2n}.$$

Set

$$\gamma_1^{(n)} := \{\zeta_0^{(n)}, \dots, \zeta_{n-1}^{(n)}\}, \quad \gamma_2^{(n)} := \{0, \zeta_0^{(n-1)}, \dots, \zeta_{n-2}^{(n-1)}\}.$$

the prototypical configurations associated respectively to $\mathcal{C}_1$ and $\mathcal{C}_2$.

Because of the repulsion, we expect that the points near the boundary are distributed close $\gamma_1^{(n)}$ and $\gamma_2^{(n)}$, so we trust the approximations

$$H(\gamma) \simeq H(\gamma_i^{(n)}), \quad \forall \gamma \in \mathcal{C}_i, \quad i \in \{0, 1\}.$$

The classical computation gives

$$H(\gamma_1^{(n)}) = -\frac{n}{2} \log n$$

and

$$H(\gamma_2^{(n)}) = -\frac{n-1}{2} \log(n-1).$$

where the impact of the central point just disappear since the zeta's are the roots of unity. In particular,

$$H(\gamma_1^{(n)}) - H(\gamma_2^{(n)}) = \frac{1}{2} \log n + \frac{1}{2} + O(n^{-1}), \quad n \rightarrow \infty.$$

Thus the free energy discrepancy is

$$\Delta_n := F_{n,\beta}(\mathcal{C}_1) - F_{n,\beta}(\mathcal{C}_2) = \frac{1}{2} \log n - \frac{2}{\beta} \log n + \frac{1}{2} + O(1).$$

Hence, for $\beta < 4$, $\Delta_n \rightarrow -\infty$ and for $\beta > 4$, $\Delta_n \rightarrow \infty$. Hence, in the first case $\mathcal{C}_2$ is more favorable while in the second case it is $\mathcal{C}_1$ that is favored.